A CMS Open Data Diagnostic Search for $H \to \tau\tau$ in the $\mu\tau_h$ Final State

Analysis my_analysis[1]

[1] *CMS Open Data reduced-sample analysis*

(Dated: June 2, 2026)

A diagnostic search for Higgs boson decays to tau pairs is performed with CMS 2012 Open Data reduced samples in the $\mu\tau_h$ final state. An optimized calibrated-score model is trained with multivariate input reweighting, widened DY/Z normalization, and a same-sign data-driven QCD/fake template. The score fit gives $\mu = 38.380^{+7.035}_{-6.116}$ from the profile likelihood and an observed 95% CLs upper limit $\mu < 50.000$, but it fails the observed validation gate. The previous visible-mass result is retained as the validated baseline: $\mu = 0.438^{+4.944}_{-0.438}$, 95% CLs $\mu < 10.764$, exp. $\mu < 10.738$.

Data and method. The analysis uses the public Run2012B/C TauPlusX HiggsTauTauReduced mirror and the CMS Open Data luminosity reference of 11.467 fb^{-1}. Signal and background simulations are normalized with official Open Data record event denominators rather than local reduced-file entries. Events are selected with the $\mu + \tau_h$ trigger, tight muon and hadronic-tau requirements including a tight anti-muon veto, opposite-sign charge, and low transverse mass.

Background correction. The W+jets normalization is measured in a high-m_T control region. The VBF-background rate is calibrated in a VBF-like top-btag control region outside the low-m_T signal region, giving a scale 0.557 ± 0.051. The reducible QCD/fake contribution is estimated from same-sign low-m_T data after subtracting non-QCD simulation. For the primary calibrated-score fit, the OS/SS transfer factor measured in the lowest score bin is 0.020 ± 0.253. The classifier input correction is derived before training from validation and control regions using a multivariate data-vs-background density-ratio model. Embedded and EWK Z reduced samples are unavailable, so the DY/Z normalization is loosened rather than replaced by fabricated samples.

Results. The primary corrected workspace has median expected 95% CLs limit $\mu < 1.974$ and observed limit $\mu < 50.000$. The best fit is $\mu = 38.380^{+7.035}_{-6.116}$ from the profile likelihood, but it fails the observed validation gate. The frozen previous visible-mass baseline gives $\mu = 0.438^{+4.944}_{-0.438}$, 95% CLs $\mu < 10.764$, exp. $\mu < 10.738$ and is retained as the validated result for this reduced Open Data workflow.

Comparison and interpretation. CMS Run 1 reported evidence for $H \to \tau\tau$ with a best-fit signal strength of 0.78 ± 0.27 and observed and expected significances of 3.2 and 3.7. CMS later observed the decay at 13 TeV with observed and expected significances of 4.9 and 4.7, and reported 5.9 standard deviations in the CMS combination. The ATLAS+CMS Run 1 combination and PDG/Higgs-summary branching fraction provide global context but are not direct validation targets for this reduced single-channel analysis. The present result is methodological reproducibility, not an independent CMS-quality evidence claim.

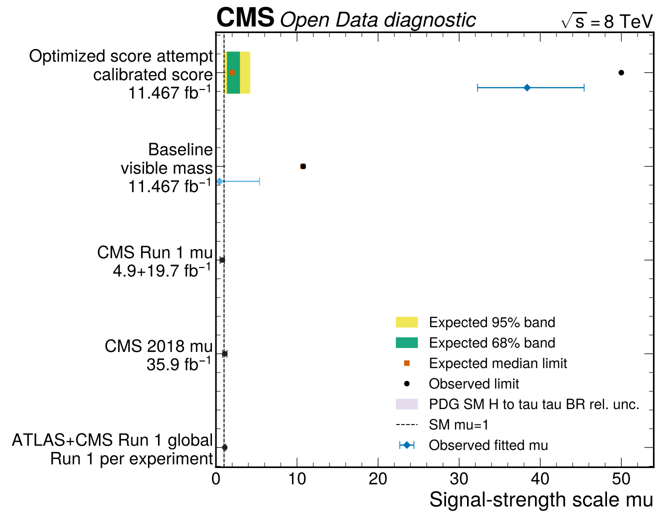

FIG. 1. Signal-strength and limit comparison. The primary open-data result is shown with expected bands and an observed marker following CMS-style limit plot conventions. The CMS/ATLAS+CMS publication rows are context, and the narrow band at $\mu = 1$ shows the PDG/Higgs-summary SM $H \to \tau\tau$ normalization uncertainty.

TABLE I. Selected yields in the primary calibrated-score fit after the QCD/fake, DY/Z, and input-reweighting corrections. Background includes DY, ttbar, W+jets, and the same-sign QCD/fake estimate.

Category	Data	Bkg.	QCD/fake	Signal
vbf	79	60.2	0.4	1.59
boosted	2213	2115.5	5.6	9.17
zero_jet	8466	14047.1	22.9	14.34

[1] CMS Collaboration, Evidence for the 125 GeV Higgs boson decaying to a pair of tau leptons, JHEP 05 (2014) 104.

[2] CMS Collaboration, Observation of the SM scalar boson decaying to a pair of tau leptons, Phys. Lett. B 779 (2018) 283.

TABLE II. Comparison of the reduced open-data results with published and world reference context. The open-data rows use only the localized 2012 $\mu\tau_h$ reduced samples and are not direct reproductions of the multi-channel CMS or ATLAS+CMS analyses.

Result	Scope	Significance	Signal-strength information
Optimized score attempt	8 TeV $\mu\tau_h$ reduced, calibrated score	12.470	$\mu = 38.380^{+7.035}_{-6.116}$, 95% CLs $\mu < 50.000$, exp. $\mu < 1.974$, gate fail
This analysis baseline	8 TeV $\mu\tau_h$ reduced, visible mass	0.095	$\mu = 0.438^{+4.944}_{-0.438}$, 95% CLs $\mu < 10.764$, exp. $\mu < 10.738$
CMS Run 1	7+8 TeV multi-channel	obs. 3.2, exp. 3.7	$\mu = 0.78 \pm 0.27$
CMS 2018	13 TeV multi-channel	obs. 4.9, exp. 4.7	$\mu \simeq 1.09$
CMS 2018 combined	CMS 7+8+13 TeV	obs. 5.9, exp. 5.9	CMS combination context
ATLAS+CMS Run 1	7+8 TeV global Higgs rates	global context	$\mu = 1.09 \pm 0.11$
PDG/HXSWG SM	$H \to \tau\tau$ branching fraction	not a search	BR about 6.3%, $\mu = 1$ convention

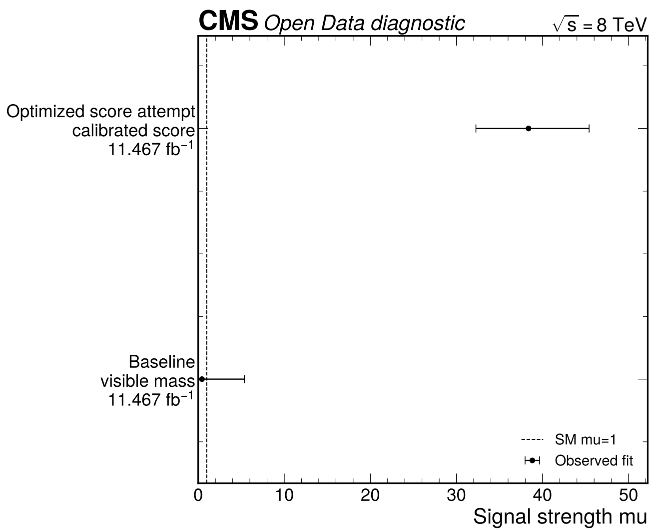

FIG. 2. Primary and baseline signal-strength comparison. The calibrated score primary result is compared with the frozen previous visible-mass baseline using profile-likelihood μ intervals.

[3] ATLAS and CMS Collaborations, Measurements of the Higgs boson production and decay rates and constraints on its couplings from a combined ATLAS and CMS analysis, JHEP 08 (2016) 045.

[4] Particle Data Group, Review of Particle Physics, Phys. Rev. D 110 (2024) 030001.

[5] CERN Open Data Portal, CMS Higgs to tau tau reduced samples for education and outreach.

[6] L. Heinrich et al., pyhf: pure-Python implementation of HistFactory statistical models.

[7] A. L. Read, Presentation of search results: the CLs technique.

[8] G. Cowan et al., Asymptotic formulae for likelihood-based tests of new physics.

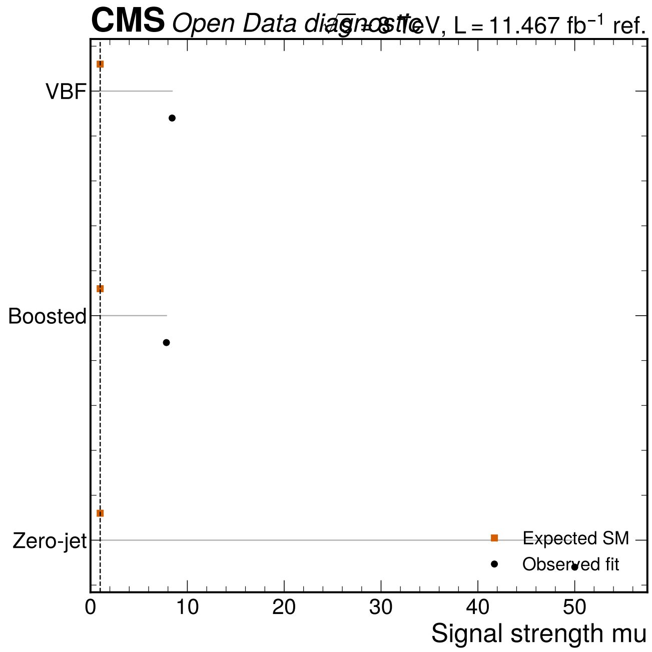

FIG. 3. Category signal-strength comparison for the primary calibrated-score model. The expected marker is the Standard Model value $\mu = 1$ and the observed marker is the single-category profile-fit value; the quoted result remains the simultaneous three-category fit.

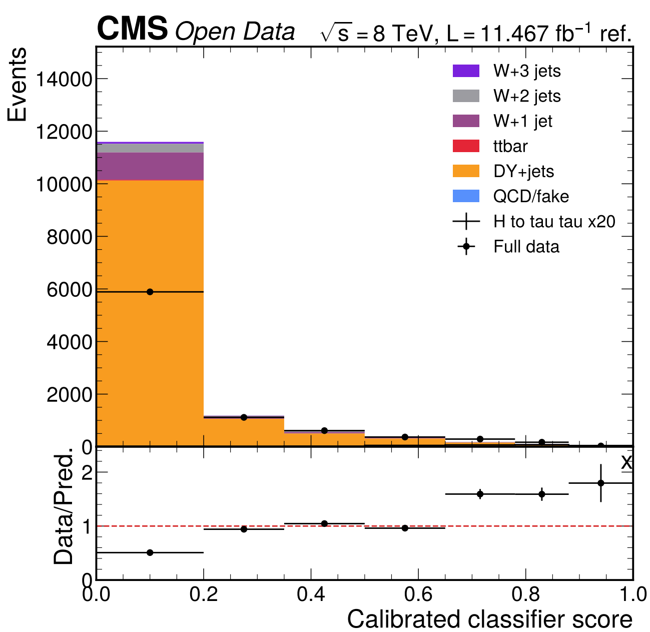

FIG. 4. Primary calibrated-score validation in the zero-jet category. The same-sign QCD/fake estimate and pre-training input reweighting stabilize the dominant zero-jet normalization.